UNIT–2 AGILE MINI-PROJECT

Student Record and Academic Management Platform

Agile/Scrum Project Planning Report

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1. Agile Project Planning Report

Project Objective: Develop a secure and user-friendly web-based platform for managing student records and academic activities including attendance, courses, marks, notifications and reports.

Agile Approach: The project follows Scrum. Development is divided into short Sprints with continuous planning, development, testing, review and improvement.

Expected Outcome: A functional platform that allows students, faculty and administrators to efficiently manage academic information.

2. Product Vision Document

Product Vision: Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application.

Primary Users: Students, Faculty, Administrators.

Business Benefits: Reduces manual record keeping; improves accuracy; saves time; provides easy access to records; simplifies academic reporting.

Success Criteria: Secure login, student record management, attendance recording/viewing, marks and grade calculation, report generation, and academic notifications.

Constraints: Limited development time, team size and resources; data security requirements; integration between modules.

3. Scrum Team Structure

Team Member	Scrum Role	Main Responsibility
Yash Dhokane	Product Owner	Requirements, stakeholder needs and backlog prioritization
Pradnya Lakariya	Scrum Master	Scrum process, ceremonies and team coordination
Shravni Thorat	Developer	Application development and implementation
Shruti Garje	Developer	Application development and testing

4. Scrum Roles & Responsibilities

Product Owner: Defines requirements, prioritizes the Product Backlog, represents stakeholder interests and accepts completed work.

Scrum Master: Facilitates Scrum ceremonies, removes impediments, ensures Agile practices are followed and supports collaboration.

Development Team: Designs the application, writes code, tests software, prepares documentation and demonstrates completed features.

5. Epic Documentation

Epic	Purpose / Major Features
Epic 1 – Student Management	Student registration, profile management, search, update and deletion
Epic 2 – Course Management	Add/update courses, assign students and view course details
Epic 3 – Attendance Management	Record attendance, view attendance and generate attendance reports
Epic 4 – Marks & Academic Performance	Enter/update marks, calculate grades and generate result sheets
Epic 5 – Authentication & Security	Login, logout, password reset and role-based access control
Epic 6 – Notifications	Attendance alerts, examination notifications and result notifications

Epic	Purpose / Major Features
Epic 7 – Reporting	Student, attendance, performance and course reports

6. User Story Repository

ID	Epic	User Story
US01	Authentication	As a user, I want to log in so that I can securely access the system.
US02	Authentication	As a user, I want to log out so that my account remains secure.
US03	Student	As an administrator, I want to add students so that their records are maintained.
US04	Student	As an administrator, I want to search students so that I can quickly find their records.
US05	Student	As an administrator, I want to update student information so that records remain accurate.
US06	Student	As an administrator, I want to delete student records so that outdated records can be removed.
US07	Course	As an administrator, I want to add courses so that courses can be managed.
US08	Course	As an administrator, I want to assign students to courses so that registrations are maintained.
US09	Attendance	As a faculty member, I want to record attendance so that attendance records remain accurate.
US10	Attendance	As a student, I want to view my attendance so that I can monitor my attendance percentage.
US11	Marks	As a faculty member, I want to enter marks so that student performance can be recorded.
US12	Marks	As a faculty member, I want to calculate grades so that academic results can be generated.
US13	Notifications	As a student, I want to receive attendance alerts so that I know when my attendance is low.
US14	Notifications	As a student, I want to receive examination notifications so that I do not miss important dates.
US15	Reporting	As an administrator, I want to generate student reports so that academic information can be analyzed.
US16	Reporting	As a faculty member, I want to generate attendance reports so that attendance can be monitored.

7. Acceptance Criteria

User Story	Acceptance Criteria
US01 – Login	Login page loads; registered users can log in; invalid credentials show an error; successful login redirects to dashboard; se
US03 – Add Student	Admin opens form; enters required details; system validates data; record is saved; new student appears in list.
US04 – Search Student	Admin enters search information; matching students are displayed; information is correct; no-result message is shown whe
US09 – Record Attendance	Faculty selects course and date; marks attendance; records save successfully; attendance report can be generated.
US11 – Enter Marks	Faculty selects student; enters marks; system validates marks; marks save successfully; updated marks can be viewed.
US12 – Calculate Grades	System retrieves marks; calculates grade; displays grade correctly; result can be included in result sheet.

8. Prioritized Product Backlog

Priority	ID	Epic	User Story
High	US01	Authentication	User Login
High	US03	Student	Add Student
High	US04	Student	Search Student
High	US09	Attendance	Record Attendance
High	US11	Marks	Enter Marks
High	US02	Authentication	Logout
Medium	US05	Student	Update Student
Medium	US07	Course	Add Course
Medium	US08	Course	Assign Students

Priority	ID	Epic	User Story
Medium	US10	Attendance	View Attendance
Medium	US12	Marks	Calculate Grades
Medium	US16	Reporting	Attendance Report
Medium	US13	Notifications	Attendance Alert
Low	US06	Student	Delete Student
Low	US14	Notifications	Examination Notification
Low	US15	Reporting	Student Report

9. Story Point Estimation

Story Points	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

ID	User Story	Points
US01	User Login	3
US02	Logout	1
US03	Add Student	5
US04	Search Student	3
US05	Update Student	3
US06	Delete Student	2
US07	Add Course	3
US08	Assign Students	5
US09	Record Attendance	5
US10	View Attendance	3
US11	Enter Marks	5
US12	Calculate Grades	5
US13	Attendance Alert	5
US14	Examination Notification	3
US15	Student Report	5
US16	Attendance Report	5

10. Sprint Planning

Sprint	Goal	Main Activities	Deliverable
Sprint 1	Authentication & Setup	Repository setup, registration, login, logout, basic UI, testing	Basic platform with working authentication
Sprint 2	Student Management	Add, search, update and delete student	Working Student Management module
Sprint 3	Course & Attendance	Add course, assign students, record/view attendance, reports	Working Course and Attendance modules
Sprint 4	Marks & Performance	Enter/update marks, calculate grades, result sheets	Working Marks and Result module

Sprint	Goal	Main Activities	Deliverable
Sprint 5	Notifications & Reporting	Attendance alerts, exam/result notifications, reports	Working notification and reporting modules

11. Sprint Backlog

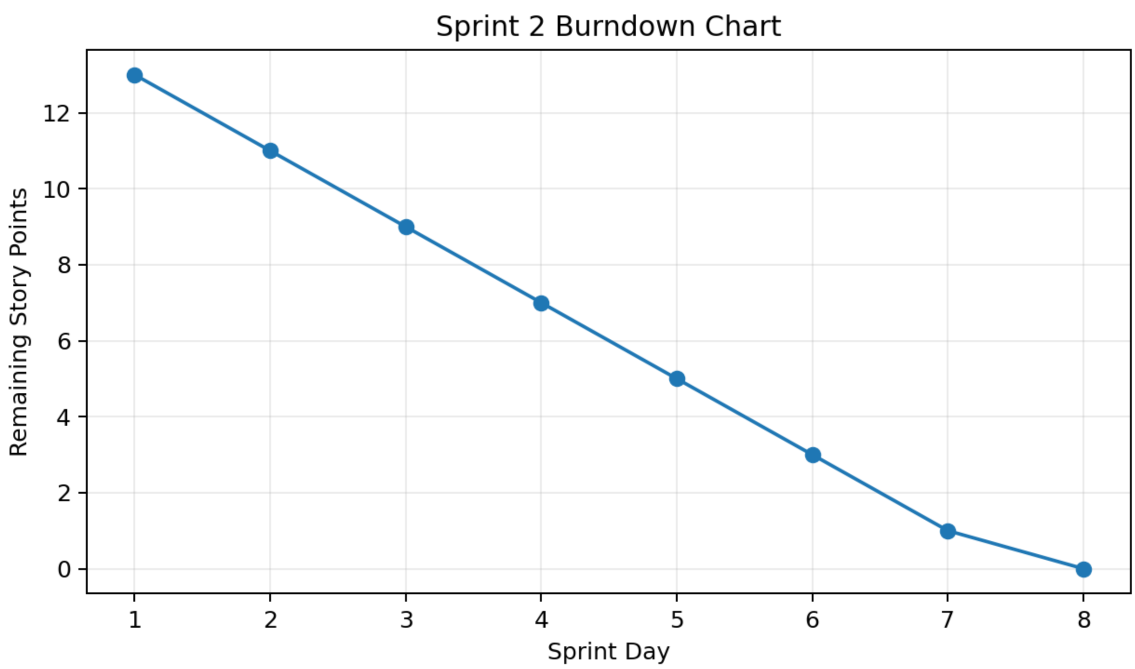
Sprint	Tasks
Sprint 1	Repository setup; Basic UI; User Registration; Login; Logout; Testing
Sprint 2	Add Student; Search Student; Update Student; Delete Student; Testing
Sprint 3	Add Course; Assign Students; Record Attendance; View Attendance; Attendance Report
Sprint 4	Enter Marks; Update Marks; Calculate Grades; Generate Result Sheets
Sprint 5	Attendance Alerts; Examination Notifications; Result Notifications; Student/Performance/Course Reports

12. Scrum Board

Backlog	To Do	In Progress	Testing	Completed
Result Sheet	Add Student	Login	—	Project Setup
Notifications	Search Student	—	—	—
Reports	Attendance	—	—	—
Course	Add Course	—	—	—

13. Burndown Chart

Example Sprint 2 remaining work: 13 → 0 story points across 8 Sprint days.



14. Scrum Ceremonies

Sprint Planning: Select User Stories, estimate effort, define Sprint Goal and create Sprint Backlog.

Daily Scrum: Each member reports what was completed yesterday, what will be done today and current challenges.

Sprint Review: Demonstrate completed features and collect feedback.

Sprint Retrospective: Discuss what went well, challenges and improvements.

15. Sprint Review Report

Sprint 1: Completed project setup, basic interface, login and registration. Feedback: keep login simple, use clear error messages and provide easy dashboard navigation.

Sprint 2: Completed Add Student, Search Student, Update Student and Delete Student. Feedback: search should be fast and student information should be validated.

16. Sprint Retrospective Report

What went well: Team collaboration, task division and achievement of Sprint objectives.

Challenges: Technical issues, module integration and longer-than-expected testing.

Improvements: Better estimation, earlier testing, improved communication and frequent code integration.

17. Agile Framework Comparison

Framework	Main Idea	Suitable For
Scrum	Sprint-based development with defined roles and ceremonies	Small/medium teams
Kanban	Visualize and continuously manage workflow	Continuous delivery
XP	Focus on engineering practices and software quality	Software development
Lean	Eliminate waste and maximize customer value	Efficient development
LeSS	Scale Scrum across multiple teams	Large Scrum projects
SAFe	Structured Agile scaling across an organization	Large enterprises
Scrum@Scale	Scale Scrum through interconnected Scrum teams	Multiple Scrum teams

18. Recommended Framework

Scrum is recommended because the team is relatively small, the application can be divided into Sprints, Scrum provides clear roles and responsibilities, features can be delivered incrementally and feedback can be collected after each Sprint.

19. Project Timeline & Release Roadmap

Phase	Sprint	Main Work
Phase 1	Sprint 1	Setup + Authentication
Phase 2	Sprint 2	Student Management
Phase 3	Sprint 3	Course + Attendance
Phase 4	Sprint 4	Marks + Academic Performance
Phase 5	Sprint 5	Notifications + Reporting
Phase 6	Final	Integration + Testing + Release

Release 1: Authentication + Basic Platform → **Release 2:** Student + Course Management → **Release 3:** Attendance + Academic Performance → **Release 4:** Notifications + Reporting → **Final Release:** Complete Platform.

20. Conclusion

The Student Record and Academic Management Platform can be effectively developed using Scrum. The project is divided into Epics, User Stories and prioritized Backlog items. Story Points estimate effort, while Sprints support incremental development. Sprint Planning, Daily Scrum, Sprint Reviews and Retrospectives help the team monitor progress, identify problems and continuously improve the product.